**331.** $25^{25}$ is divided by 26, the remainder is
(*a*) 1 (*b*) 2
(*c*) 24 (*d*) 25

**332.** If $(67^{67} + 67)$ is divided by 68, the remainder is
(*a*) 1 (*b*) 63
(*c*) 66 (*d*) 67

**333.** One less than $(49)^{15}$ is exactly divisible by
(*a*) 8 (*b*) 14
(*c*) 50 (*d*) 51

**334.** The remainder when $7^{84}$ is divided by 342 is
(*a*) 0 (*b*) 1
(*c*) 49 (*d*) 341

**335.** The remainder when $2^{60}$ is divided by 5 equals
(*a*) 0 (*b*) 1
(*c*) 2 (*d*) 3

**336.** By how many of the following numbers is $2^{12} - 1$ divisible?
2, 3, 5, 7, 10, 11, 13, 14
(*a*) 4 (*b*) 5
(*c*) 6 (*d*) 7

**337.** The remainder when $(15^{23} + 23^{23})$ is divided by 19, is
(*a*) 0 (*b*) 4
(*c*) 15 (*d*) 18

**338.** When $2^{256}$ is divided by 17, the remainder would be
(*a*) 1 (*b*) 14
(*c*) 16 (*d*) None of these

**339.** $7^{6n} - 6^{6n}$, where $n$ is an integer > 0, is divisible by
(*a*) 13 (*b*) 127
(*c*) 559 (*d*) All of these

**340.** It is given that $(2^{32} + 1)$ is exactly divisible by a certain number. Which of the following is also definitely divisible by the same number?
**(S.S.C., 2007)**
(*a*) $2^{16} + 1$ (*b*) $2^{16} - 1$
(*c*) $7 \times 2^{33}$ (*d*) $2^{96} + 1$

**341.** The number $(2^{48} - 1)$ is exactly divisible by two numbers between 60 and 70. The numbers are
**(A.A.O. Exam, 2010)**
(*a*) 63 and 65 (*b*) 63 and 67
(*c*) 61 and 65 (*d*) 65 and 67

**342.** $n$ being any odd number greater than 1, $n^{65} - n$ is always divisible by
(*a*) 5 (*b*) 13
(*c*) 24 (*d*) None of these

**343.** Let $N = 55^3 + 17^3 - 72^3$. Then, $N$ is divisible by
(*a*) both 7 and 13 (*b*) both 3 and 13
(*c*) both 17 and 7 (*d*) both 3 and 17

**Directions (Questions 344–345):** *These questions are based on the following information:*
Given $N = \underline{|1} + \underline{|2} + \underline{|3} + \ldots\ldots + \underline{|99} + \underline{|100}$.

**344.** Find the last two digits of $N$.
(*a*) 00 (*b*) 13
(*c*) 19 (*d*) 23

**345.** Find the remainder when $N$ is divided by 168.
(*a*) 33 (*b*) 67
(*c*) 129 (*d*) 153

**346.** What is the remainder when $4^{61}$ is divided by 51?
(*a*) 20 (*b*) 41
(*c*) 50 (*d*) None of these

**347.** What is the remainder when $17^{36}$ is divided by 36?
(*a*) 1 (*b*) 7
(*c*) 19 (*d*) 29

**348.** Which one of the following is the common factor of $(47^{43} + 43^{43})$ and $(47^{47} + 43^{47})$?
(*a*) $(47 - 43)$ (*b*) $(47 + 43)$
(*c*) $(47^{43} + 43^{43})$ (*d*) None of these

**349.** Find the product of all odd natural numbers less than 5000.
(*a*) $\dfrac{5000!}{2500 \times 2501}$ (*b*) $\dfrac{5000!}{2^{2500} \times 2500!}$
(*c*) $\dfrac{5000!}{2^{5000}}$ (*d*) None of these

**350.** How many zeros will be required to number the pages of a book containing 1000 pages?
(*a*) 168 (*b*) 184
(*c*) 192 (*d*) 216

**351.** If $a^2 + b^2 + c^2 = 1$, what is the maximum value of $abc$?
(*a*) $\dfrac{1}{3}$ (*b*) $\dfrac{1}{3\sqrt{3}}$
(*c*) $\dfrac{2}{\sqrt{3}}$ (*d*) 1

**352.** Find the unit's digit in the sum of the fifth powers of the first 100 natural numbers.
(*a*) 0 (*b*) 2
(*c*) 5 (*d*) 8

**353.** If the symbol $[x]$ denotes the greatest integer less than or equal to $x$, then the value of
$$\left[\frac{1}{4}\right] + \left[\frac{1}{4} + \frac{1}{50}\right] + \left[\frac{1}{4} + \frac{2}{50}\right] + \ldots\ldots + \left[\frac{1}{4} + \frac{49}{50}\right] \text{ is}$$
(*a*) 0 (*b*) 9
(*c*) 12 (*d*) 49

**354.** When $100^{25} - 25$ is written in decimal notation, the sum of its digits is
(*a*) 444 (*b*) 445
(*c*) 446 (*d*) 448

**355.** What is the number of digits in the number $(1024)^4 \times (125)^{11}$?
(*a*) 35 (*b*) 36
(*c*) 37 (*d*) 38